MIL-S-4083C
11 October 1974

SUPERSEDING
MIL-S-4083B
13 February 1964

MILITARY SPECIFICATION

SAW, RECIPROCATING,

PORTABLE, PNEUMATIC: METAL-CUTTING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of pneumatic, portable, reciprocating, metal-cutting saw.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-P-40	- Packaging and Packing of Hand Tools.

Military

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-C-4109	- Coupling Assembly, Low Pressure, Air Hose, Quick-Disconnect.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

FSC 5130

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MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1474	- Noise Limits for Army Materiel.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF LABOR

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, and Amendments - Occupational Safety and Health Administration Standards.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The saw shall consist essentially of an air motor, reciprocating assembly, housing, handle, and blade chuck. The air motor shall be either reciprocating or rotary type.

3.2 First article (preproduction model). Unless otherwise specified the supplier shall furnish one or more saws, for examination and testing within the time frame specified (see 6.2), to prove prior to starting production that his production methods and choice of design detail will produce saws that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examination and tests shall be conducted by the supplier subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests as specified (see 6.2).

3.3 Material. Material shall be as specified herein. Material not specified shall be new and of a quality necessary to produce saws to meet the requirements specified herein.

3.4 Handle. The saw shall be provided with a pistol-grip-type handle located to permit one-hand operation of the saw. The handle shall be tapped with 1/4-inch NPT to receive the air coupling nipple.

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3.4.1 Control lever. An instant-release-type, air-control lever or trigger shall be provided and located in the handle to permit operation of the saw and the air-control lever at the same time with one hand. The control lever shall provide infinite speed control from stop to full rated speed.

3.4.2 Air-coupling nipple. An air-coupling nipple conforming to MIL-C-4109, Type II, Class A, Style 1, Size 1/4-inch, shall be provided.

3.5 Blade and file chuck. A blade and file chuck shall be provided and mounted on the end of the reciprocating shaft. The chuck shall securely hold a heavy duty hack-saw blade 1/2 inch wide and 0.050 inch thick. The chuck shall prevent loosening or loss of the blade during sawing operation by chucking, clamping, or keying the blade in position. The chuck or a separate, interchangeable chuck shall lso mount and secure files having round shanks up to 1/4 inch in diameter or flat shanks in sizes up to 13/16 inch by 5/32 inch.

3.6 Stroke. The length of the stroke shall be not more than 1-3/4 inches or less than 1/4 inch at the no-load speed. The cutting action shall be on the return stroke of the blade.

3.7 Health and safety. A guard shall be provided for the reciprocating shaft to protect the operator and to provide a guide and rest during sawing and filing operations. Guards and other appropriate protection shall be provided for the operator of the saw in accordance with appropriate occupational safety and health regulations.

3.7.1 Sound level requirement. The sound level of the saw shall be in accordance with MIL-STD-1474, Category D, for steady-state noise when tested in accordance with 4.5.2.2.4. Hearing protectors need not be supplied with the saw. Hazard signs, if required, shall state: "CAUTION, HEARING PROTECTION REQUIRED."

3.8 Housings. The housing of the handle motor and reciprocating mechanism shall be of nonferrous material. The reciprocating housing shall be fitted with a bronze bushing splined or flatted to receive the reciprocating shaft. The housings shall be buffed and polished and shall be free of all machine-grinding and filing marks.

3.9 Dimensions. The overall length of the saw shall be not more than 14-1/2 inches, excluding the blade.

3.10 Weight. The weight of the saw, excluding the blade, shall be not more than 5-1/2 pounds.

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3.11 Performance. The saw shall cut 1/4-inch 6061-T6 aluminum plate at an average rate of not less than 3 inches per minute at the applicable cutting strokes per minute and air consumption as specified in Table I.

Table I. Performance Characteristics

Type air motor	Strokes per minute (minimum)	Air consumption (maximum) (cfm)
Reciprocating	900	15.0 at 85 ± 5 psi
Rotary	750	25.0 at 85 ± 5 psi

3.12 Air motor.

3.12.1 Reciprocating type. The reciprocating air motor shall be double-acting and shall transmit power directly to the reciprocating shaft.

3.12.1.1 Reciprocating shaft. The reciprocating shaft shall be attached to the air motor to form the reciprocating mechanism. The shaft shall have splines or flats to prevent rotation in the housing.

3.12.2 Rotary air motor. The rotary air motor shall be of the multivane type and shall transmit power to the reciprocating shaft through a rotary cam and positive acting cam follower. Springs shall not be used to effect the return or cutting stroke.

3.12.2.1 Reciprocating mechanism. The reciprocating mechanism shall consist of a rotary cam, cam follower, and reciprocating shaft. Cam follower and reciprocating shaft shall convert the rotary action of the air motor to reciprocating action for the saw blade. The rotary cam shall be supported at each end with sealed-for-life bearings. The reciprocating shaft shall have splines or flats to prevent rotation in the housing.

3.13 Lubrication. The air motor shall be lubricated through a standard air line lubricator. The reciprocating mechanism (see 3.12.2.1), shall be provided with a means of flood lubrication of the cam and cam follower.

3.14 Identification marking. The saw shall be identified in accordance with MIL-STD-130. Identification plates shall conform to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a conspicuous protected location.

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3.19 Workmanship. All parts, components, and assemblies of the saw including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, lint, fins, pits, sprues, scale, flux, and other harmful extraneous materials. All edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection of the saw (see 4.3).
(b) Quality conformance inspection of the saw (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection. Prior to the examination and test, the saw shall be completely lubricated with military oils. Oils shall be those designated for use in the ambient temperature at which the test is being performed.

4.3.1 Examination. The saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The saw shall be tested as specified in 4.5.2.2. Failure to pass any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4.2 Tests. Each saw shall be tested as specified in 4.5.2.2. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The saws shall be examined for the following defects:

101. Material not as specified.
102. Handle not as specified.
103. Control valve not as specified.
104. Coupling nipple not as specified.
105. Blade and file chuck not as specified.
106. Stroke not as specified.
107. Assembly incomplete or incorrect.
108. Dimension and weight not as specified.
109. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to performing the tests, the saw shall be serviced with military lubricants designated for use at the ambient temperature at which the test is being conducted. An air-line lubricator, filled with lubricant, shall be connected in the air line.

4.5.2.2 Performance tests.

4.5.2.2.1 Air consumption. Connect the saw to a flow meter. Open the air-control valve and, while maintaining an air pressure of 85 psi plus or minus 5 psi at the air inlet connection, determine the air consumption. Air consumption in excess of 15.0 cfm for the reciprocating type motor or 25 cfm for the rotary air motor shall constitute failure of this test.

4.5.2.2.2 Free speed. Operate the saw with the air-control valve wide open, maintaining an air pressure of 85 psi plus or minus 5 psi at the air inlet connection. Determine the free speed with a strobe light. A free speed of less than 600 cutting strokes per minute for the reciprocating air motor or less than 750 cutting strokes per minute for the rotary motor shall constitute failure of this test.

4.5.2.2.3 Sawing. Secure a suitable piece of 1/4-inch-thick 6061-T6 aluminum in a vise. Make three cuts for 1 minute each. Failure of the saw to maintain an average cutting rate of 3 inches per minute shall constitute failure of this test.

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3.15 Workmanship. All parts, components, and assemblies of the saw including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, lint, fins, pits, sprues, scale, flux, and other harmful extraneous materials. All edges shall be rounded or beveled.

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4.3.1 Examination. The saw shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The saw shall be tested as specified in 4.5.2.2. Failure to pass any test shall be cause for rejection.

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4.5.2.1 Test conditions. Prior to performing the tests, the saw shall be serviced with military lubricants designated for use at the ambient temperature at which the test is being conducted. An air-line lubricator, filled with lubricant, shall be connected in the air line.

4.5.2.2 Performance tests.

4.5.2.2.1 Air consumption. Connect the saw to a flow meter. Open the air-control valve and, while maintaining an air pressure of 85 psi plus or minus 5 psi at the air inlet connection, determine the air consumption. Air consumption in excess of 15.0 cfm for the reciprocating type motor or 25 cfm for the rotary air motor shall constitute failure of this test.

4.5.2.2.2 Free speed. Operate the saw with the air-control valve wide open, maintaining an air pressure of 85 psi plus or minus 5 psi at the air inlet connection. Determine the free speed with a strobe light. A free speed of less than 600 cutting strokes per minute for the reciprocating air motor or less than 750 cutting strokes per minute for the rotary motor shall constitute failure of this test.

4.5.2.2.3 Sawing. Secure a suitable piece of 1/4-inch-thick 6061-T6 aluminum in a vise. Make three cuts for 1 minute each. Failure of the saw to maintain an average cutting rate of 3 inches per minute shall constitute failure of this test.

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4.5.2.2.4 Noise. The saw shall be placed in a test area which is free of any large reflecting surfaces, such as signs or hillsides, within 50 feet of the saw or sound-level measuring equipment. With the saw in use, sound level measurements shall be made adjacent to the saw in accordance with MIL-STD-1474 and with the recording microphones located to simulate the ear position of an operator using the tool. Conformance with the noise level criteria specified in 3.7.1 shall constitute successful completion of this test. If the criteria are not met, the minimum distance from the tool at which the noise levels are acceptable shall be determined, and that distance shall be included in the test report (see 6.6).

4.6 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the saws shall be examined and tested to determine compliance with the Quality Assurance Provisions of PPP-P-40.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. The saws shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. Preservation and packaging shall be Level A or C, and packing shall be Level A, B, or C, as specified (see 6.2).

6. NOTES

6.1 Intended use. The saw covered by this specification is intended for general utility maintenance usage to perform various operations in conjunction with sheet metal work.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) When a preproduction model is not required (see 3.2).
(c) Time frame required for submission of the preproduction model and number of saws to be furnished (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the supplier (see 3.2).
(e) Level of preservation and packaging and level of packing required (see 5.1).

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6.3 Preproduction model. Any changes or deviations of production saws from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish saws conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and suppliers maintenance and operation manual(s) to be furnished with each saw.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of saws.

6.6 Ear protection. When the noise level exceeds Category D of MIL-STD-1474, the operator's manual for the saw shall include appropriate instructions for, (1) use of hearing protection, and (2) distance from the saw at which 85dB(A) will be met.

Custodians:

Army - ME
Navy - SH
Air Force - 84

Review activity:

Army - GL

User activity:

Navy - YD

Preparing activity:

Army - ME

Project No. 5130-0216

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